

WE ARE SEEING THE BEGINNINGS OF A RAPIDLY CHANGING PLANET

THE KEELING CURVE

IN CONTEXT

KEY FIGURE

Charles Keeling (1928–2005)

BEFORE

1896 The Swedish chemist Svante Arrhenius is the first to estimate the extent to which atmospheric CO_2 could increase Earth's temperature.

1938 Comparing historic temperature data and CO_2 measurements, British scientist Guy Stewart Callendar concludes that the increase in CO_2 is responsible for the warming of the atmosphere.

AFTER

2002 The European Space Agency's ENVISAT satellite begins to produce up to 5,000 readings of greenhouse gases every day.

2020s Data from NASA's Orbiting Carbon Observatory-3 detects a dip in CO_2 levels over Los Angeles during 2019–21, when people stop using their cars in the Covid-19 pandemic.

The Keeling Curve, named after Charles Keeling, an American scientist, charts the daily record of atmospheric carbon dioxide (CO_2), measured in parts per million by volume (ppmv), in a series dating back to 1958. It